Sahaj Bhadja

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Summary

Software integration engineer with 2+ years building **Linux-based automation**, **hardware-software validation workflows**, and **C++/Python tooling** for mission-critical systems, fleet configuration, and rapid test iteration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **Python orchestration services** and operational interfaces for 500+ users, reducing processing latency from **1.2s to 350ms** while improving real-time observability and fault isolation.
- Implemented **validation safeguards and rollback controls** for 5K+ monthly jobs, improving deployment reliability across distributed system components and high-change integration cycles.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **testable backend/frontend integration features** handling 10K+ daily requests, translating requirements into reusable APIs and stable modules for production release paths.
- Improved **CI/CD release automation** and defect triage workflows, reducing turnaround by **60%** while maintaining quality across rapidly iterating product environments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **simulation-aligned evaluation pipelines** for 100K+ trajectories, enabling reproducible testing, anomaly detection, and disciplined benchmark execution in safety-critical scenarios.
- Produced **technical diagnostics and system summaries** to bridge research and implementation teams, accelerating debugging handoffs and integration decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API/ETL integration services** with Python, AWS Glue, and S3, improving response times by **30%** while supporting reliable cross-system data exchange pipelines.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito on Linux/Windows environments, increasing pipeline efficiency by **68%** and reducing issue recurrence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **MERN + WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving synchronization reliability and operational consistency for distributed users.
- Automated **Docker/Jenkins deployment workflows**, cutting deployment time by **50%** with repeatable script-driven build, validation, and rollback-ready release processes.

Skills

Languages: **Python**, **C++**, Java, JavaScript/TypeScript, SQL, Bash
Integration & Systems: Linux/UNIX, hardware-software checkout workflows, DAQ-style telemetry handling, anomaly triage
Networking: TCP/IP fundamentals, routing/subnetting basics, service and environment configuration
Build & Ops: Docker, Jenkins, GitHub Actions, CI/CD, Make/CMake, automated testing (PyTest, Mockito)
Engineering: Git/GitHub, Agile/Scrum, cross-functional coordination, technical documentation

Projects

Hardware-in-the-Loop Telemetry Harness | *Python, C++, Linux, CAN*
- Built a **HIL-style integration harness** for sensor/actuator simulation with deterministic replay and fault injection, improving software checkout speed and traceability during integration testing.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLFQ** with runtime metrics and visualization tooling, reducing debugging time by **20%** through repeatable test-driven scheduling analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized processing API** with automated deployment validation on AWS, improving reliability and uptime across production-grade operational data workloads.